<u>Variation of nitrogen oxides in Mile End and Reading.</u>

Abstract

This report will detail how, over the years from 2000 to 2024, nitrogen oxides (NOx) varied in a dense and populated area, such as Mile End, and a less dense area, such as Reading. Using this data, it was concluded that Reading had less pollution than Mile End but as time went on, Mile End's NOx pollution did see a decrease which could be explained by London's introduction of the Low Emission Zone or Ultra Low Emission Zone (ULEZ) which had a profound impact in decreasing pollution.

Arthur Luong
Student ID: 240264996
Department of Physics and Astronomy, School of Physical and Chemical Sciences, Queen Mary University of London, London, E1 4NS.
Date of submission: 17/04/2025

Introduction

Nitrogen oxides pollution industrially comes from the combustion of natural gas, fuels, and the manufacturing of nitrogen fertilisers. The combustion of natural gas contributes the most to NOx emissions. However, with the combustion of fuels, it could contribute up to 50% of the NOx emissions. Although contributing the least, the manufacturing of nitrogen fertilisers does become prominent in areas of low NOx pollution as the other two sources will not be as dominant as in areas of moderate to high NOx pollution [1].

This project will use data from Londonair, which is the website of the London Air Quality Network (LAQN) and shows the type of air pollution and amount of pollution in specific parts of London or close to London [2]. For this project, it will focus on the NOx pollution from Mile End and Reading. This project will also include the method used to analyse the data and the conclusions gathered from the data.

Method

For this project, data from Mile End and Reading spanning from 2000 to 2024 was from the LondonAir project, which is downloadable online.

First, the data from the weekdays and weekends were averaged separately per month by doing:

$$Average = \frac{Sum\ of\ all\ the\ data\ from\ weekdays\ or\ weekends\ in\ the\ month}{Amount\ of\ weekdays\ or\ weekends\ within\ the\ month}. \tag{1}$$

Repeating this for every month in the span of 2000 to 2024 for both Mile End and Reading, to then plot four lines on a NOx pollution against date graph to get a visible variation.

Second, the project involved zooming in on a set of times for which further analysis would be carried out. For this project, using the plot from the first step, the years 2012 to 2018 were chosen to be the area of analysis, as they contained the fewest zero values, as those values would not make a fair comparison since there was no data within that date range. Using this, the percentage from 1st January 2012 could be calculated using:

$$Percentage\ change\ =\ \frac{Yearly\ average - 1st\ January\ 2012\ data}{1st\ January\ 2012\ data} \times 100. \quad (2)$$

By plotting a percentage change against the year graph, more conclusions could be gathered.

Results

Using the method described above, the first graph looks like:

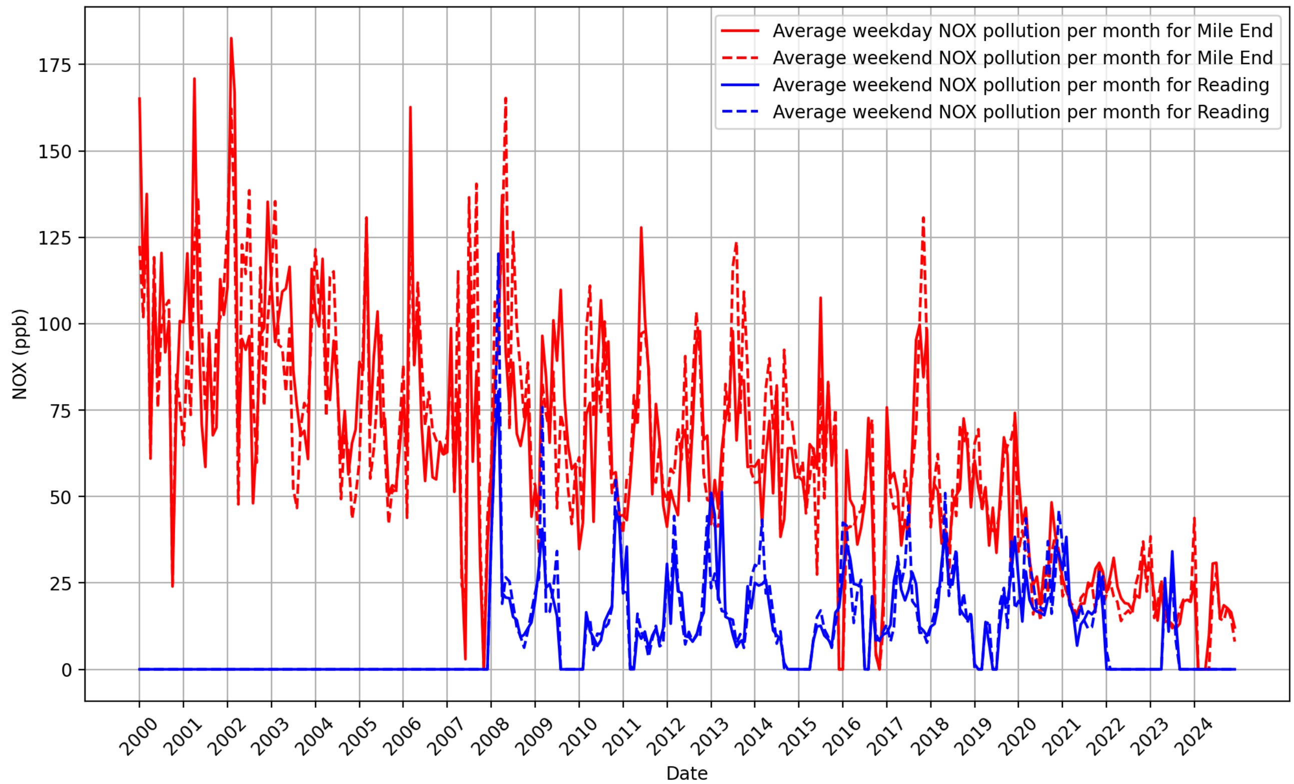

Figure 1: The variation of NOx pollution against time with the weekdays and weekends from both Mile End and Reading.

Figure 1 shows that Mile End tends to have higher pollution than Reading, which is plausible as Mile End is in London, which is much more populated than Reading, so it would also be plausible to assume there is more fuel being burnt due to vehicles.

The figure also shows that in 2020 and 2021, Mile End's NOx emissions dropped sharply. This can be explained by London's introduction to the Ultra Low Emission Zone (ULEZ), where European emissions standards were enforced [3] through charging vehicles that did not meet the emissions requirement, which essentially decreased NOx emissions.

Although Reading is not in London, the plot shows that NOx pollution for Reading decreased too, which may be because of the European emissions standards between Euro 3 and Euro 6 itself, where car models were strictly manufactured to reduce their NOx emissions [4] when the United Kingdom was in the European Union and post Brexit.

The figure also shows Mile End steadily decreasing in NOx pollution and almost matching Reading. This may also be due to the European emissions standards, as it was implemented in 2001.

By plotting the second graph, it looks like:

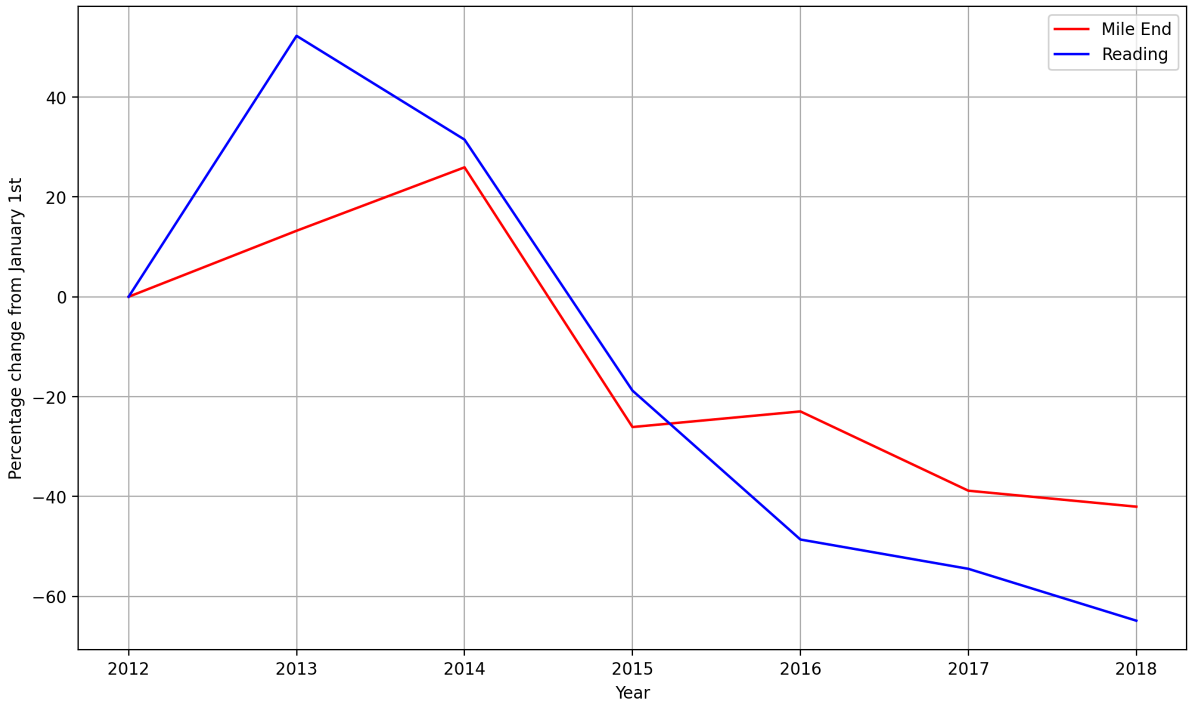

Figure 2: Percentage change of yearly average NOx pollution from January 1st 2012, from both Mile End and Reading.

Here, there was a large increase in 2013 for Reading and 2014 Reading, where from the start of 2012, NOx pollution rose by 50% in 2013 for Reading and more than 20% in Mile End. This may be due to the rise in diesel cars, which contributed to the rise in NOx pollution [5] [6].

As with the first graph, there is a decrease in NOx pollution after 2014. This may be due to not only the European emissions standards but also the Volkswagen emissions scandal, where the emission tests for their cars were falsified, which then led to a decrease in popularity for diesel cars as a whole [7], which then led to a decrease in NOx pollution.

Conclusions

In conclusion, because of the European emissions standards, the NOx pollution decreased for populated and vehicle-heavy areas such as Mile End and even decreased for areas outside of London, like Reading. Moreover, after 2020, the expansion of ULEZ has led to further decreases in NOx pollution in Mile End.

The plots could be better if Mile End or Reading did not have points where there was no data for those dates, as it may be an unfair comparison of the two sites, whilst also limiting a full analysis of the data between 2000 and 2024.

Therefore, this report showed that as time went on, NOx pollution decreased due to new or ongoing policies being implemented as although only Mile End and Reading were analysed, it still showed that there will be cleaner air across the United Kingdom with time.

References

[1] Kumar, S., (2002). *How is NOx Formed.* Energy Solutions Centre, http://cleanboiler.org/workshop/how-is-nox-formed/

[2] LondonAir., *About LondonAir*, n.d., https://www.londonair.org.uk/LondonAir/General/about.aspx

[3] Transport For London., *Ways to meet the standard*, n.d., https://tfl.gov.uk/modes/driving/ultra-low-emission-zone/ways-to-meet-the-standard?intcmp=52221

[4] RAC., (2025). *Euro 1 to Euro 7 guide – find out your vehicle's emissions standard*., https://www.rac.co.uk/drive/advice/emissions/euro-emissions-standards/

[5] Rosenbaum, M., (2017). *Why officials in Labour government pushed 'dash for diesel'*. BBC, https://www.bbc.co.uk/news/uk-politics-41985715

[6] Franco, V., Sánchez, F. P., German, J., Mock, P., (2014). *Real-world exhaust emissions from modern diesel cars.* International Council on Clean Transportation, https://theicct.org/publication/real-world-exhaust-emissions-from-modern-diesel-cars/

[7] Topham, G., Clarke, S., Levett, C., Scruton, P., Fidler, M., (2015). *The Volkswagen emissions scandal explained*. The Guardian, https://www.theguardian.com/business/ng-interactive/2015/sep/23/volkswagen-emissions-scandal-explained-diesel-cars